STAGE 2 – PRACTICAL ENGINEERING CHALLENGE

TEAM ID:	TEAM NAME:
LRC-26-0260	Aftershock

Problem Statement: Industrial Manufacturing Process Automations

Industrial manufacturing depends on highly controlled liquid-handling systems to deliver process fluids reliably from a source or bulk supply to downstream process equipment. Depending on the application, these fluids may include **ultrapure fluids, process chemicals, solvents, developers, or CMP-related fluids**. Industry standards explicitly address liquid-chemical distribution, high-purity piping, bulk chemical distribution, system qualification, and UPW system operation and monitoring.

In such systems, **flow is a fundamental process variable**. An incorrect flow rate can indicate or cause a restriction, incorrect valve state, inadequate pumping performance, abnormal process demand, or another hydraulic deviation. The consequence may be unstable process conditions, incorrect chemical delivery, equipment malfunction, or loss of process performance.

This challenge therefore focuses on a fundamental industrial facilities and process-engineering task: **To establish a defined process-fluid flow, measure it reliably, detect a deviation, diagnose its cause, and restore the process to the required operating condition.**

Engineering Challenge

As you are part of a multidisciplinary engineering team responsible for designing, building and validating a small-scale automated liquid-flow system represented of industrial process handling system.

CASE 1 – CONTROL + MEASUREMNT

Problem: Develop a system that manages the changing level of fluid in the sump automatically (when there is a sudden external fluid charging to the sump) and while accurately measuring and totalizing the fluid discharged from the sump using the flow transmitter.

CASE 2 – FLUSHING +SHUTDOWN

Problem: Develop an alternative operating configuration that allows flush water to pass through the system while all three pumps operate simultaneously, and then safely transition the system to an OFF condition when the flush-water supply has been exhausted.

Given Process Parameters

Pipe diameter =	**8mm**
Fluid density (ρ) =	**1000kg/m^3**
Dynamic Viscosity (μ) =	**1x10^{-3}Pa.s**
Required Flow range =	**1-6 L/min**
Nominal target flow for the live challenge =	**3L/min**

Available Hardware

Mechanical Hardware	Electronic Hardware
1. Water Tank – 3 Quantity	1. Custom PCB
2. Pumps – 3 Quantity	2. ESP32 NodeMCU module
3. FT-01 Flow Transmitter – 01 Quantity	3. Pump Driver-1
4. Ball valve – 3 Quantity	4. Pump Driver-2
5. ON/OFF/XV Valve – 1 Quantity	5. Pump Driver-3
6. 10-mm OD tubing/blue – 9 meters/1 meter	6. 12 V DC power input (**SMPS**)
7. Microcontroller/PCB/ data-acquisition software	7. 12 VDC Solenoid-valve interface
8. Required electrical/ electronic accessories	8. Flow-sensor interface
	9. Float-switch interface
	10. ON/OFF-switch interface
	11. Pump-driver PWM/enable interfaces
	12. On/Off Switch

The team must determine the appropriate arrangement, connections and measurement method.

- Assemble and commission the liquid-flow system using the components provided.
- Measure and characterize the flow using FT-01
- Interface FT-01 with the provided **data-acquisition system**
- Experimentally determine the relationship between sensor output and actual flow.
- Characterize the effect of valve position on flow rate. Record and analyse flow vs time data
- Maintain target flow of 3L/min
- Use graphs and measured data to justify engineering decisions
- Design an appropriate hydraulic arrangement and P&ID

TEAM INSTRUCTIONS

- As a team go through the list provided to read about each component that has been Listed.
- The PCB is a 12 V DC and the SMPS connections should be made carefully.

TASK 1: HYDRAULICS SYSTEM DESIGN

Q1.1 Design and build the hydraulic path with the flow transmitter on the recirculation line. Submit/sketch a labelled P&ID below by showing the pumps, Needle valves, flow transmitter, flow direction, Tank, sump, tube diameter and measurement points.

CASE 1 – Process Fluid Transfer & Automatic Level Protection

Objective: Set-up and operate a closed-loop process- fluid handling system using the available tank, pump, valves flow transmitter, sump, float sensor and control PCB. The System must continuously transfer process fluid recirculation system by connecting the 10L fluid tank with a On/Off valve to a pump-01 and the pump-01 goes to the inlet of sump 4.2L tank. The Sump 4.2L outlet line is connected to another pump-02 followed by a flow transmitter that measures the total flow back to the 10L fluid tank. The Pump-01 runs on 50% duty cycle and Pump-02 runs on 100% duty cycle. **Calibrate the flow transmitter** and demonstrate correct fluid circulation, if there is sudden introduction of external fluid into the Sump tank then Pump-01 should stop while pump – 02 should still runs till fluid level reduces in the sump. After the fluid reduction in the SUMP 4.2L tank the Pump-01 should turn on and the process must continue. **Also, demonstrate Flow vs Time trends in mL/s, total transferred Liters back to the 10L tank** (share Screenshots flow in L/min). **Share the code**. The case-1 can be turned on using the available on/off Switch, to start the system.

CASE 2 – Flush/Line Conditioning Sequence

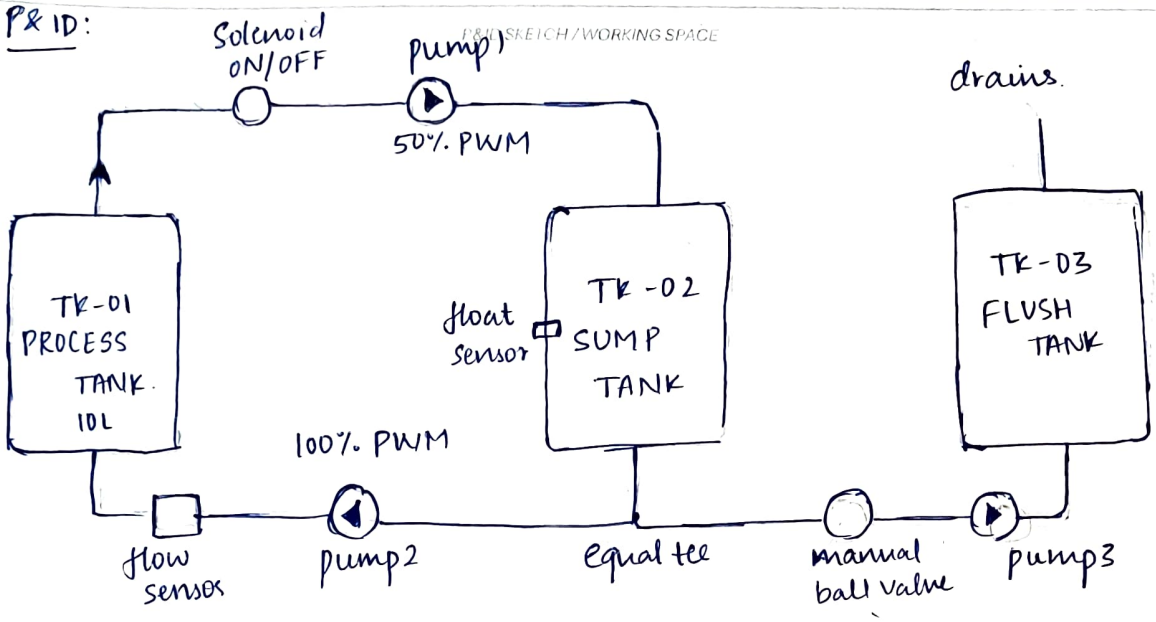
Objective: After successfully completing case 1, initiate the flush water in the sequence using the **equal Tee.** The On/Off valve must be switched off in order to isolate the 10L process fluid tank. The flush Fluid tank (4.2L) is used as a temporary fluid source to flush out the process fluid or any debris from the system. Before the start of this process make sure you open the manual ball valve on the fluid flush line before Pump-03 starts. The entire process of case -1 and case - 2 should run in a single go without manual intervention except opening of ball valve before the Pump-03 starts. **The System must stop as soon as the tank flush fluid 4.2L tank finishes.** The system must be automated without human interventions (only ball valve can be opened manually before the start of flush line sequence)

SUBMIT:

A. Submit: 10-15 min Video and Photograph of the completed setup, with identification of each component and explain the direction of flow in the video verbally. Demonstrate both Case - 1 and Case – 2.

B. Final P&ID Block diagram

P&ID:



→ All the pumps are connected to PCB

Case 1 Working:

1. Solenoid valve opens, and pump1 runs at 50% PWM to transfer fluid from the 10L tank into 4.2L tank.
2. Pump2 runs at 100% duty cycle, draining the sump through Flow transmitter back into the 10L Tank.
3. When external fluid makes the sump full, the Float switch triggers HIGH, immediately shutting pump1 OFF
4. pump2 continues running at 100% to drain the sump, once the level drops back to normal pump1 turns ON automatically.

Case 2 Working:

1. When case2 starts solenoid is CLOSED, isolating 10L tank.
2. pump2 runs alone at 100% with manual ball valve opened, added line impedance causes the flow at FT-01 to drop from 2.72 L/min.
3. When the manual ball valve is opened, added line impedance causes the flow at FT-01 to drop from 2.72 L/min down to 2.66 L/min.
4. Detecting flow below 2.72 L/min for 2 seconds, the ESP32 automatically start Pump3 at 100%.
5. pump3 draws clean water from 4.2L Flush Tank through the Equal Tee, flushing process fluid & debris through the sump.
6. Float switch is ignored here, pump2 & pump3 run continuously until liquid is evacuated i.e. measured flow drops to 0.00 L/min (<0.60 L/min).

Q1.2 After setting-up the system, now operate it at approximately at 1L/min, 3L/min & 6L/min. Compare the theoretical velocity with the actual operating behaviour.

WORKING / COMPARISON

Theoretical Fluid Velocity Formula: From the hydraulic continuity equation for incompressible flow in a pipe.

$$Q = A \cdot V \Rightarrow V = \frac{Q}{A} = \frac{4Q}{\pi d^2}$$

parameters:

$Q \rightarrow 1\,L/min = 1.667 \times 10^{-5}\,m^3/s$

$d \rightarrow 10mm = 0.010m$

$A \rightarrow \frac{\pi d^2}{4} = \frac{\pi (0.010)^2}{4} = 7.854 \times 10^{-5}\,m^2$

v (Theoretical Velocity): Fluid velocity in m/s.

$Re = \frac{Vd}{\nu}$

Calculations:

Case 1: $Q = 1L/mm$

$Q = 1.667 \times 10^{-5}\,m^3/s$

$V = \frac{1.667 \times 10^{-5}}{7.854 \times 10^{-5}} = 0.212 m/s$

$Re = \frac{0.212 \times 0.010}{1 \times 10^{-6}} \approx 2,120$

Case 2: $Q = 3L/min$

$Q = 5 \times 10^{-5}\,m^3/s$

$V = \frac{5 \times 10^{-5}}{7.854 \times 10^{-5}} = 0.637 m/s$

$Re = \frac{0.637 \times 0.010}{1 \times 10^{-6}} = 6,370$

Case 3: $Q = 6L/min$

$Q = 1 \times 10^{-4}\,m^3/s$

$V = \frac{1 \times 10^{-4}}{7.854 \times 10^{-5}} = 1.273 m/s$

$Re = \frac{1.273 \times 0.010}{1 \times 10^{-6}} = 12,730$

Actual operating vs Theoretical Velocity

1. At Low Flow (≈1L/min):
- Setup: Throttling the manual ball valve to 50% or operating at 50% PWM
- Observation: Fluid moves smoothly with minimal acoustic noise & negligible pipe vibration
- Deviation from theory: At v = 0.21m/s, the fluid velocity is close to the lower threshold of turbine. Internal mechanical friction on the rotor's pin causes slight measurment lag & non-linearity close to 1L/min

2. At Normal Flow (≈3L/min)
- Setup: Pump2 running at 100% duty cycle.
- Observation: Continuous, stable
- Theory Comparsion: Kinetic energy of flow easily overcomes rotor friction, allowing sensor to operate in its most linear & accurate range.

3. At High Flow (≈6L/min):
- Limitation: A single pump is unable to achieve 6L/min. 6L/min is delivered at/when two pumps are running in parallel.
- Friction: Since Darcy-Weisbach head loss scales with v^2, hydraulic resistance is 36 times > at 1L/min
- Observation: This rapid suction velocity can induce inlet vortexing & cavitation bubbles, which cause fluctuations in flow readings

TASK 2: FLOW RESPONSE TO VALVE POSITION

Q 2.1 Set the ball valve to approximately 25%,50%,75% & 100%. At each position, allow the flow to stabilize and record the flow rate. Create the following graph of valve position (%) vs Flow Rate (L/min)

VALVE POSITION	FLOW RATE
25%	0.0 L/min
50%	1.130 L/min
70%	2.593 L/min
100%	2.791 L/min

GRAPH

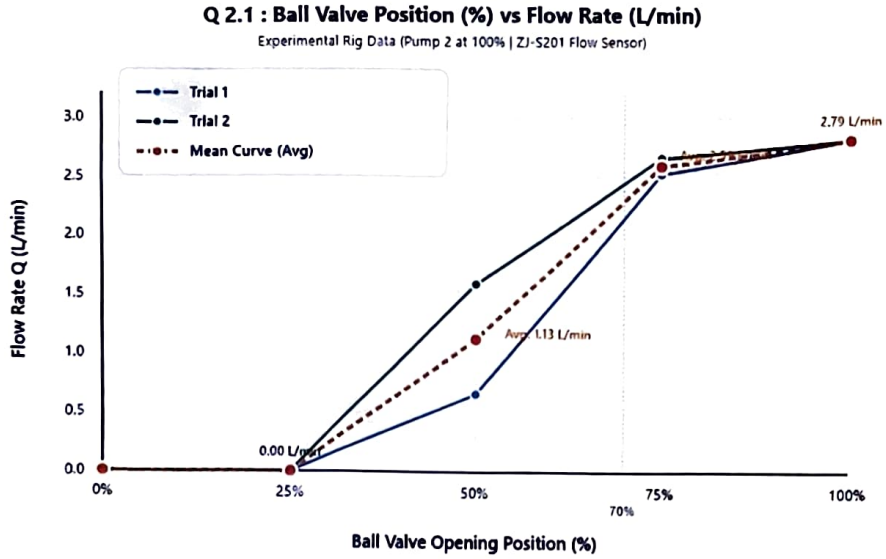
Q2.2 Determine the experimental calibration equation: $Q = af + b$, where Q = flow rate in L/min, f = pulse frequency in Hz, a = experimental calibration coefficient and b = offset.

Report:

R^2 =	0.999
a =	0.1329 L/min·Hz
b =	0.0003 L/min
Maximum calibration error =	0.0005 L/min (at 2Hz)
Average calibration error =	0.0003 L/min

Q2.3 FT-01 produces pluses corresponding to water flow. Run the system at least for 5 different flow conditions within the available operating range (Hint: use formulas f=N/t and Q=V/t). For each condition, record the following:

Trial	Pulses	Time(s)	Frequency (Hz)	Collected Volume (L)	Actual Flow (L/min)
1	0 (0% open)	1	0.0	0.000	0.000
2	(25% open) 0	1	0.0	0.000	0.000
3	(50% open) 5	1	5.0	0.011	0.665
4	(75% open) 20	1	20.0	0.044	2.659
5	(100% open) 21	1	21.0	0.046	2.791



x-axis : Valve position (%) - 0%, 25%, 50%, 75%, 100%
y-axis : Flow rate (L/min) - 0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0 L/min

In 2.2) $Q = 0.1329 * f + 0.0003$ [L/min]

In 2.3) $f = \frac{N}{t}$ and $Q = \frac{V}{t} \times 60$

Calibration factor : 451.389 pulses/L

$V = \frac{N}{451.3}$ (Total collected volume)

TASK 3: HYDRAULIC SYSTEM THEORY AND FAULT ANALYSIS

Q3.1 The effective liquid volume inside a section of tubing is 0.5L, while the operating flow rate is 2.5L/min. The hydraulic residence time is given by $T_r = Q/V$. Calculate the residence time in seconds, also answer the following questions:

1. What does residence time physically represent in this system?
2. What happens to residence time if the tubing volume is doubled while the flow rate remains constant?
3. What happens to residence time if the flow rate is doubled while the tubing volume remains constant?
4. Why is residence time important when interpreting a sudden change in measured flow?

ANSWER Given,

$V = 0.5\,L$

$Q = 2.5\,L/min = 0.04167\,L/s$

Calculation:

$$T_r = \frac{V}{Q} = \frac{0.5\,L}{2.5\,L/min} = 0.2\,minutes$$

1. Hydraulic residence time represents the average time taken by a packet of fluid to travel through the tubing section from the inlet to the outlet. In this rig, it is the physical transport delay required for fluid leaving the pump to travel through the pump to travel through the tubing before reaching the flow sensor or sump.
2. The residence time doubles because from $T_r = \frac{V}{Q}$, residence time is directly proportional to volume. Doubling the fluid capacity of the tube at the same volumetric flow rate means twice as much liquid must be displaced, requiring twice as long.
3. The residence time halves because from $T_r = \frac{V}{Q}$, residence time is inversely proportional to flow rate. Doubling the flow rate doubles fluid velocity clearing the same tube volume in half the time.
4. Residence time introduces a physical transpostation lag into the system.
   1. Delayed Sensor Response: A sudden upstream change (such as opening the manual ball valve or turning on a pump) is not reflected at the sensor instantaneously. The system requires at least one residence time for the velocity & pressure wavefronts to reach steady state across tube length.
   2. Preventing Premature Control Triggers: Automated control logic must include a stabilization delay (such as the 8-12 s wait in our control code) that matches or exceeds the hydraulic residence time. This prevents false sensor readings caused by air bubbles or transient flow spikes during fluid transit.

Q3.2 The following observation are obtained from the same system:

CONDITION	PUMP	VALVE	FT-01 PULSE SIGNAL	MEASURED FLOW
A	ON	Open	Normal	Normal
B	ON	Partially closed	Reduced	Reduced
C	ON	Open	Zero	Zero
D	OFF	Open	Zero	Zero
E	ON	Open	Intermittent	Fluctuating

For each of the condition, identify the most likely physical or instrumentation condition. Then answer: Which condition is most difficult to distinguish using FT-01 alone, and what additional information would you require to distinguish the fault? (Hint: Teams must consider possibilities such as: Valve restriction, pump failure, hydraulic blockage, sensor/signal failure, unstable flow etc)

Identification of Physical / Instrumentation conditions :
Condition A → The system is fully primed, unobstructed, & running under nominal design
condition B → The partially closed valve introduces hydraulic head loss, causing a proportional decrease in fluid velocity & pulse frequency.
condition C → No flow or complete signal loss.
Condition D: → The pump is intentionally de-energized, without a driving head, fluid is stationary, and zero pulses is the correct, expected output.
condition E → Unstable Flow or cavitation or signal noise

2. Most difficult to distinguish using FT-01 alone would be condition C. because in, condition C, FT-01 output 0Hz (zero pulses) despite the pump being commanded ON, The flow transmitter alone cannot determine whether
→ Water is not moving because the pump failed mechanically / electrically.
→ Water is not moving due to physical blockage or dry tank,
→ Water is actually flowing, but the sensor / wiring has failed
All these completely different physical faults yield the identical reading : 0.00L/min

3. To isolate the exact root cause in C
pump motor current :
•0A Current : Electrical fault
• Abnormally High current : pump is jammed
• Normal : pump motor is running normally, fault either a downstream blockage, empty tank or sensor failure.
In-line pressure transmitter :
•High Discharge pressure + zero flow : Confirms a hydraulic blockage downstream
•Zero : Confirms pump failure or loss of prime suction line.
Tank liquid level sensor : Verifies if liquid is physically available at the suction port, ruling out a dry-run fault.
Sensor Electrical Line Check : If line voltage is stuck low or disconnected, it confirms an instrumentation rather than a hydraulic issue.

HARDWARE LIST

Mechanical Hardware

Component	Description	Main Use
Ball Valve (BV-01)	A manually operated valve with a spherical shut-off mechanism.	Used to **manually isolate or permit fluid flow** in the pipeline. It is useful for maintenance, testing, and flow isolation.
10-Litre Fluid Tank (Fluid Tank-01)	Transparent storage tank with approximately 10 L capacity.	Used as the **main fluid reservoir** for storing process fluid. The transparent body also allows the fluid level and condition to be visually monitored.
5-Litre Sump	Small transparent reservoir labelled **SUMP**, approximately 4.5 L.	Used as a **collection/storage vessel** for fluid returning to or entering the sump section of the system.
4.2-Litre Flush Fluid Tank / Flush Line	The transparent vessel labelled **FLUSH FLUID** is approximately 4.5 L.	Provides **flush fluid** for cleaning or flushing the main pipe circuit when required.
ON/OFF Valve (OV-01)	Electrically operated valve/solenoid-type valve shown in the setup.	Used for **automatic control of fluid flow**, allowing the controller to turn a particular flow path on or off.
Pumps -Pump-1, Pump-2, Pump-3	Motor-driven fluid pumps. Three pumps are shown in the setup.	Used to **circulate and transfer fluid** between different sections of the system. Each pump can be independently controlled.
8 mm Process Pipe	Flexible blue tubing with approximately 8 mm process-line size.	Used as the **main fluid-flow path** connecting tanks, valves, pumps, tees, and other components.
10 mm × 8 mm Reducers	Reducer fittings that transition between a 10 mm connection and an 8 mm pipe.	Used to **adapt the pump connection size to the 8 mm process pipe. Mainly used for Pumps**
8 mm Connectors	Push-fit/barbed-type fittings compatible with the 8 mm tubing.	Used to **join pipes to components** and to make detachable connections between sections of tubing.
8 mm Equal Tee	T-shaped connector having three equal 8 mm ports.	Used to **split one 8 mm fluid line into two branches or combine two branches into one line.**

Electronic Hardware

Electronic Component	Identification / Function	Use
PCB -Printed Circuit Board	Custom control board	Provides the electrical interconnection between the **12 V power supply, ESP32 controller, sensors, valves, indicators, and pump-driver interfaces.**
ESP32 / NodeMCU ESP32S module (U7)	Main microcontroller	Acts as the **central controller.** It provides GPIO/PWM signals for controlling outputs and receiving signals from sensors/switches.

Electronic Component	Identification / Function	Use
12 V DC Input Connector	Marked **12V INPUT**	Connection point for the external 12 V DC power supply.
12 V Output Connectors	P1, P2, P3 and additional 12 V output terminals	Provide 12 V power connections for external loads such as the pump-driver/motor circuits.
Motor Driver Interfaces -U2, U3, U4	Three multi-pin driver connectors shown on the PCB	Provide dedicated interfaces for **Pump-1, Pump-2 and Pump-3 driver/control connections**.
Solenoid Valve Connector	Labelled **SOLENOID_VALVE -S/V/G**	Interface for controlling/connecting the solenoid or ON/OFF valve.
Flow Sensor Connector	Labelled **FLOW_SENSOR - S/V/G**	Interface for the water-flow sensor; supplies power and receives the sensor signal. **THE FLOW SENSOR SHOULD BE CALIBRATED**
Float Switch Connector	Labelled **FLOAT_SWITCH -FSW/GND**	Interface for a float/level switch used to detect a water-level condition.
ON/OFF Switch Connector	Labelled **ON/OFF_SWITCH -SW/GND**	Interface for an external manual ON/OFF switch.
12V DC SMPS	**12V DC SMPS**	For Power Supply to the board